

Problem 5

Problem 5

Problem 1 Evaluate each of the following limits. Possible answers include a number, $+\infty$, $-\infty$ and “Does Not Exist” (DNE). Make sure to state the form of the limit. Justify your answer.

(a) $\lim_{x \rightarrow 3^-} \frac{x^2 - 3}{x^2 - x - 6}$

Explanation. $\lim_{x \rightarrow 3^-} \frac{x^2 - 3}{x^2 - x - 6} = \lim_{x \rightarrow 3^-} \frac{x^2 - 3}{(x - 3)(x + 2)}$ is of the form $\frac{\#}{0}$.

Since $\lim_{x \rightarrow 3^-} (x^2 - 3) = 6$, and $\lim_{x \rightarrow 3^-} (x^2 - x - 6) = \lim_{x \rightarrow 3^-} (x - 3)(x + 2) = 0$, and since

$$x^2 - x - 6 = (x - 3)(x + 2) < 0, \text{ if } x < 3 \text{ and } x \text{ close to } 3.$$

$$\lim_{x \rightarrow 3^-} \frac{x^2 - 3}{x^2 - x - 6} = -\infty$$

(b) $\lim_{x \rightarrow 5^+} \frac{x^2 + 6}{x^2 - 3x - 10}$

Explanation. The limit is of the form $\frac{\#}{0}$.

$$\lim_{x \rightarrow 5^+} \frac{x^2 + 6}{x^2 - 3x - 10} = \lim_{x \rightarrow 5^+} \frac{x^2 + 6}{(x - 5)(x + 2)} = \infty, \text{ because}$$

$\lim_{x \rightarrow 5^+} (x^2 + 6) = 31$ and $x^2 - 3x - 10 = (x - 5)(x + 2) > 0$, for $x > 5$ and x close to 5.

(c) $\lim_{x \rightarrow 1} \frac{4 - x}{x^2 - 2x + 1}$

Explanation. The limit is of the form $\frac{\#}{0}$.

Checking left and right sided limits we see:

$$\lim_{x \rightarrow 1^+} \frac{4 - x}{x^2 - 2x + 1} = \lim_{x \rightarrow 1^+} \frac{4 - x}{(x - 1)^2} = \infty$$

and

$$\lim_{x \rightarrow 1^-} \frac{4-x}{x^2-2x+1} = \lim_{x \rightarrow 1^-} \frac{4-x}{(x-1)^2} = \infty$$

$$\text{Since } \lim_{x \rightarrow 1^-} \frac{4-x}{x^2-2x+1} = \lim_{x \rightarrow 1^+} \frac{4-x}{x^2-2x+1} \implies \lim_{x \rightarrow 1} \frac{4-x}{x^2-2x+1} = \infty$$

$$(d) \lim_{x \rightarrow 2} \frac{-e^x}{(2-x)^3}$$

Explanation. The limit is of the form $\frac{\#}{0}$.

Checking right limit we have:

$$\lim_{x \rightarrow 2^+} \frac{-e^x}{(2-x)^3} = \infty$$

and checking the left limit we have:

$$\lim_{x \rightarrow 2^-} \frac{-e^x}{(2-x)^3} = -\infty$$

$$\text{Since } \lim_{x \rightarrow 2^+} \frac{-e^x}{(2-x)^3} \neq \lim_{x \rightarrow 2^-} \frac{-e^x}{(2-x)^3} \implies \lim_{x \rightarrow 2} \frac{-e^x}{(2-x)^3} \text{ Does not exist}$$

$$(e) \lim_{x \rightarrow 1} \frac{\sin x}{\sqrt{2-x^2}-1}$$

Explanation. The limit is of the form $\frac{\#}{0}$.

Checking right limit we have:

$$\lim_{x \rightarrow 1^+} \frac{\sin x}{\sqrt{2-x^2}-1} = -\infty,$$

$\lim_{x \rightarrow 1^+} \sin x = \sin 1 > 0$, since $1 < \frac{\pi}{2}$. Explanation: Note that for x near 1 and such that $x > 1$, we have that

$x^2 > 1$, and by multiplying by (-1) , we get

$-x^2 < -1$, and by adding 2 on both sides, we get

$2-x^2 < 2-1$, and by taking the square root, we get

Problem 5

$$\sqrt{2-x^2} < \sqrt{1}, \text{ or}$$

$$\sqrt{2-x^2} < 1, \text{ and by subtracting 1 from both sides, we get}$$

$$\sqrt{2-x^2} - 1 < 0.$$

Therefore the numerator is positive, and denominator is **negative** and goes to 0.

$$\text{Checking the left limit we have: } \lim_{x \rightarrow 1^-} \frac{\sin x}{\sqrt{2-x^2}-1} = +\infty$$

Explanation: Note that and for x near 1 such that $x < 1$, we have that

$$x^2 < 1, \text{ and by multiplying by } (-1), \text{ we get}$$

$$-x^2 > -1, \text{ and by adding 2 on both sides, we get}$$

$$2-x^2 > 2-1, \text{ and by taking the square root, we get}$$

$$\sqrt{2-x^2} > \sqrt{1}, \text{ or}$$

$$\sqrt{2-x^2} > 1, \text{ and by subtracting 1 from both sides, we get}$$

$$\sqrt{2-x^2} - 1 > 0.$$

Therefore the numerator is positive, and denominator is **positive** and goes to 0.

Since $\lim_{x \rightarrow 1^+} \frac{\sin x}{\sqrt{2-x^2}-1} \neq \lim_{x \rightarrow 1^-} \frac{\sin x}{\sqrt{2-x^2}-1} \implies \lim_{x \rightarrow 1} \frac{\sin x}{\sqrt{2-x^2}-1}$ does not exist.